

Exploring Aligned and Disaligned Multimodal Features in Recommender Systems

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Outline

- 1 Introduction
- 2 Aligned Models
- 3 Experimental Settings
- 4 Experiments
- 5 Discussion & Conclusion

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Recommender Systems

- Guide users to relevant items in a large space [5]
- Approaches: Content-Based, Collaborative, Hybrid, Knowledge-Based, Context-Aware

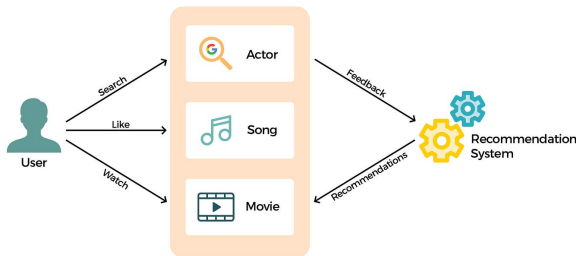


Figure: Example of a Recommender System

Multimodal Recommender Systems

- Use multiple modalities (text, image, audio) to enrich item representations [7]
- Encoding options: concat, fusion/attention, GNNs [1, 4]



Figure: Example of multimodal data for a movie (Star Wars)

Motivation & Problem Statement

Traditionally, MMRec systems extract features using independent encoders (e.g., BERT for text, ResNet for images). In other domains, like Few-Shot Learning for Image Classification, aligned features (CLIP) have shown superior performance with respect to disaligned ones.

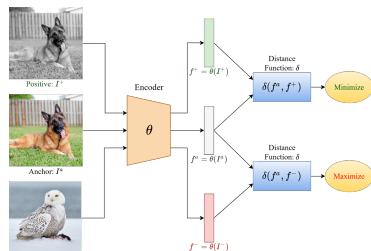
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Contrastive Learning

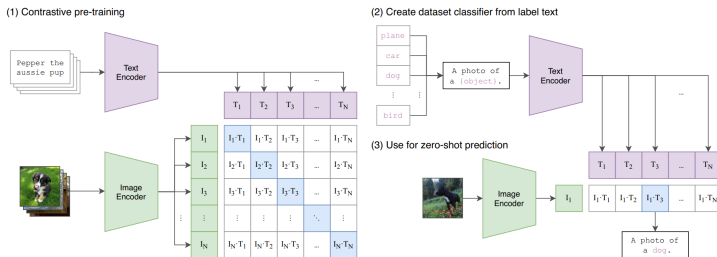
- Self-supervised method: learns representations that bring similar samples closer [3]
- Goal:** Minimize distance between positive pairs, maximize distance between negative pairs.
- Contrastive loss:

$$\ell_{i,j} = -\log \frac{\exp(\text{sim}(z_i, z_j)/\tau)}{\sum_{k=1}^{2N} [k \neq i] \exp(\text{sim}(z_i, z_k)/\tau)}$$



CLIP (Contrastive Language-Image Pre-training)

- Aligns image and text embeddings in a shared space (OpenAI) [6]
- Relevance:** Large multimodal encoders like CLIP can improve performance when fine-tuned end-to-end [8]



AudioCLIP

- Extends CLIP by adding an **audio encoder** (ESResNeXt) [2]
- **Training process:**
 - 1 Audio-head pre-training
 - 2 Cooperative distillation (aligning audio to frozen CLIP space)
 - 3 Full tri-modal fine-tuning on AudioSet

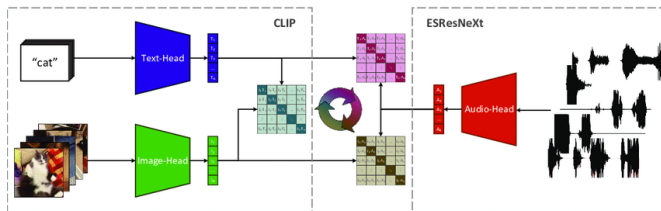
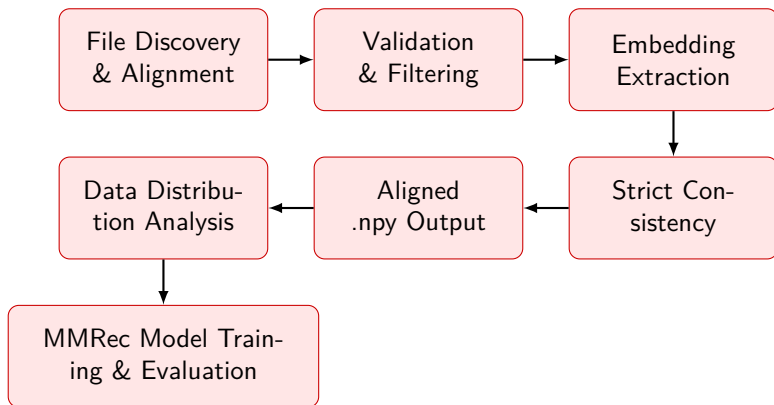


Figure: Tri-modal alignment in AudioCLIP

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Pipeline



Research Questions

RQ1: Qualitative Analysis (Geometry)

How does the alignment process (e.g., CLIP, AudioCLIP) affect the geometric distribution of the feature space?

RQ2: Quantitative Analysis (Performance)

Can the integration of *aligned* multimodal embeddings (text, image, audio) enhance recommendation accuracy compared to unimodal or non-aligned approaches?

Datasets Overview

Challenge: Dealing with different structural levels of multimodal data.

Feature	MovieLens 1M	LastFM
Domain	Movies	Music
Interactions	User-Movie (Explicit Rating)	User-Artist (Implicit Listen)
MM Level	Item Level (Movie)	Track Level (Sub-Item)
Modalities	Posters, Trailers, Plots	Audio, Covers, Metadata
Pipeline	Direct Mapping	Artist-to-Track Expansion

Table: Comparison of datasets and processing strategies

Models Evaluated

We compared performance across simpler projection models and complex Graph Neural Networks:

- **BPR (Baseline):** Bayesian Personalized Ranking (ID-only).
- **VBPR:** Visual BPR extended to support multimodal features (projection-based).
- **LATTICE:** GNN that mines latent structures between items.
- **FREEDOM:** GNN with modality-specific encoders and attention fusion.

Metrics Evaluated:

- Recall@k
- NDCG@k
- Precision@k
- MAP@k

Experimental Settings:

- **Cut-offs (k):**
 $k \in \{5, 10, 20, 50\}$
- **Filtering:** 5-core filtering applied to ensure interaction density.

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Embedding Distribution Analysis (UMAP + KDE)

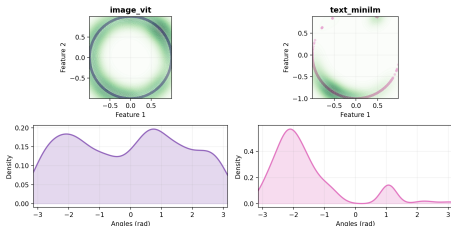
To understand *why* models perform differently, we analyzed the geometric structure of the embeddings.

- **Technique:** UMAP projection to 2D $\rightarrow$ S1 Normalization (Unit Circle) $\rightarrow$ Gaussian KDE.
- **What to look for:**
 - *Uniformity:* Do features cover the space?
 - *Peaks:* Distinct peaks indicate specific feature orientations.
 - *Collapse:* Overlapping distributions imply loss of modality-specific signal.

MovieLens 1M: Aligned vs Disaligned

Disaligned (MiniLM+ViT)

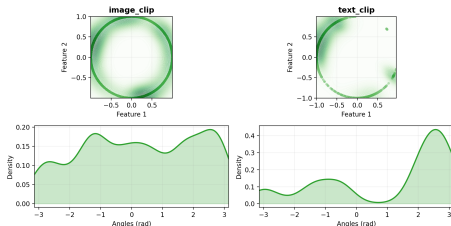
MOVIELENS_1M - text_minilm_image_vit



Distinct feature spaces preserve complementary info.

CLIP Aligned (Text+Image)

MOVIELENS_1M - text_clip_image_clip



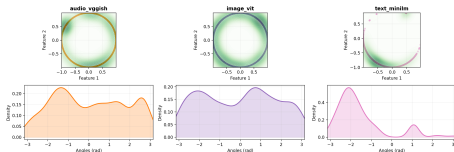
Semantic alignment bridges modalities without collapse.

- **Disaligned:** Features maintain specific orientations, capturing different aspects of user preference (e.g., visual style vs. narrative).
- **CLIP Aligned:** Successfully brings Text and Image closer semantically while maintaining enough structural diversity to be discriminative (Best Performance).

The "AudioCLIP Anomaly": Signal Collapse

Disaligned (Rich Signal)

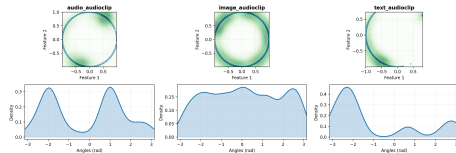
MOVIELENS_1M - text_minilm_image_vit_audio_vggish



Modality-specific distributions cover the manifold.

AudioCLIP (Over-Aligned)

MOVIELENS_1M - audioclip_full



Aggressive alignment.

Why does alignment fail here?

- **Loss of Nuance:** By forcing all three modalities (Text, Image, Audio) into the exact same space too aggressively, the model loses the rich multimodal signal

Quantitative Results: MovieLens 1M (k=10)

VBPR with CLIP features generally outperforms others. AudioCLIP underperforms.

Model	Rec@10	NDCG@10	Prec@10	MAP@10
BPR	0.1776	0.2159	0.1631	0.1120
VBPR (CLIP)	0.1911	0.2300	0.1745	0.1211
VBPR (AudioCLIP)	0.1570	0.1922	0.1476	0.0969
VBPR (Disaligned)	0.1876	0.2271	0.1718	0.1195
LATTICE_3 (CLIP)	0.1877	0.2275	0.1721	0.1198
LATTICE_3 (Disaligned)	0.1848	0.2257	0.1710	0.1183
FREEDOM (CLIP)	0.1796	0.2211	0.1657	0.1158
FREEDOM (AudioCLIP)	0.1844	0.2259	0.1688	0.1193

Table: MovieLens 1M results. **VBPR (CLIP)** achieves state-of-the-art.

Quantitative Results: LastFM (k=10)

In music domain, disaligned features preserve distinct signals better.

Model	Rec@10	NDCG@10	Prec@10	MAP@10
BPR	0.1132	0.1534	0.1404	0.1061
VBPR (CLIP)	0.1222	0.1646	0.1488	0.1149
VBPR (Disaligned)	0.1307	0.1755	0.1560	0.1242
LATTICE_3 (CLIP)	0.1232	0.1667	0.1507	0.1160
LATTICE_3 (Disaligned)	0.1258	0.1722	0.1535	0.1209
FREEDOM (Disaligned)	0.0986	0.1322	0.1224	0.0954

Table: LastFM results. **VBPR (Disaligned)** outperforms aligned models.

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Discussion: Context Matters

MovieLens 1M (Movies)

- Modalities (Posters, Plot) share high semantic correlation.
- **Result:** Alignment (CLIP) helps. It reinforces the shared signal.

LastFM (Music)

- Modalities (Audio, Cover Art) are semantically distinct/orthogonal.
- **Result:** Strict alignment hurts. Disaligned features (keeping specific audio/visual signals separate) perform better.

Conclusions and Future Work

- **Aggressive Alignment (AudioCLIP):** Can be detrimental if it collapses the feature space too much, erasing fine-grained details needed for ranking.
- **Simplicity Wins:** Simple projection models (VBPR) with high-quality features often beat complex GNNs (LATTICE) on these datasets.
- **Future Directions:**
 - Develop domain-adaptive fusion strategies (soft-alignment).
 - Investigate regularization techniques to prevent feature collapse in tri-modal alignment.

Thank you!

Thank you for your attention!



References I



Xi Chen, Yangsiyi Lu, Yuehai Wang, and Jianyi Yang.
Cmbf: Cross-modal-based fusion recommendation algorithm.
Sensors, 21(16), 2021.



Andrey Guzhov, Federico Raue, Jörn Hees, and Andreas Dengel.
Audioclip: Extending clip to image, text and audio.
In *ICASSP 2022-2022 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 976–980. IEEE, 2022.



Haigen Hu, Xiaoyuan Wang, Yan Zhang, Qi Chen, and Qiu Guan.
A comprehensive survey on contrastive learning.
Neurocomputing, 610:128645, 2024.

References II



Peishan Li, Weixiao Zhan, Lutao Gao, Shuran Wang, and Linnan Yang.

Multimodal recommendation system based on cross self-attention fusion.

Systems, 13(1), 2025.



Pasquale Lops, Marco de Gemmis, and Giovanni Semeraro.

Content-based Recommender Systems: State of the Art and Trends, pages 73–105.

Springer US, Boston, MA, 2011.

References III



Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, et al.

Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pages 8748–8763. PmLR, 2021.



Giuseppe Spillo, Elio Musacchio, Cataldo Musto, Marco de Gemmis, Pasquale Lops, and Giovanni Semeraro.

See the movie, hear the song, read the book: Extending movielens-1m, last.fm-2k, and dbbook with multimodal data.

In *Proceedings of the Nineteenth ACM Conference on Recommender Systems, RecSys '25*, page 847–856, New York, NY, USA, 2025. Association for Computing Machinery.

References IV



Zixuan Yi, Zijun Long, Iadh Ounis, Craig Macdonald, and Richard Mccreadie.

Large multi-modal encoders for recommendation.

arXiv preprint arXiv:2310.20343, 2023.